

DRAFT

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Question #: 1

A 24-year-old immunocompromised man is being treated for a chronic abscess in the ischioanal fossa. On further investigation, another large abscess was identified located on the anterior surface of the psoas major. On physical examination, there was loss of the cremasteric reflex. Which of the following areas will have reduced sensation due to the location of this abscess?

- A. Anterior aspect of the scrotum
- ✓B. Skin over the femoral triangle
- C. Antero-lateral thigh
- D. Hypogastric region of the abdominal wall
- E. Superior aspect of the gluteal region

Rationale: Correct: Nerve compressed is Genitofemoral nerve, which can be seen splitting into genital and femoral branches as it travels along psoas muscle. Genital branch: supplies cremaster muscle and medial scrotum; femoral branch: supplies skin of the femoral triangle.

Incorrect: sensation to Antero-lateral thigh is supplied by Lateral femoral cutaneous nerve. Anterior aspect of the scrotum and labia majora is supplied by Ilioinguinal nerve. Hypogastric (suprapubic) region of the abdominal wall is supplied by the iliohypogastric nerve.

Question #: 2

Following an emergency appendectomy a patient who complained of having numbness (paraesthesia) of the skin in the suprapubic region of the abdomen. The most likely nerve(s) that has been injured during the operation is:

- A. Genitofemoral
- ✓B. Iliohypogastric
- C. Subcostal
- D. Lateral femoral cutaneous and iliohypogastric
- E. Ilioinguinal and iliohypogastric

F. Ilioinguinal

Rationale: Correct: Iliohypogastric supplies internal oblique-transversus muscles, skin of the suprapubic region of the abdominal wall and superior lateral aspect of the gluteal region

Incorrect : Genitofemoral nerve , which can be seen splitting into genital and femoral branches as it travels along psoas muscle. Sensation to Antero-lateral thigh is supplied by Lateral femoral cutaneous nerve. Anterior aspect of the scrotum and labia majora is supplied by Ilioinguinal nerve.

Question #: 3

A 30-year-old man comes to the ER with excruciating abdominal pain. The on-call surgeon decides to perform an exploratory laparotomy. Where would the incision most likely be made to separate the left and right rectus sheaths?

- A. Midaxillary line
- ✓B. Linea alba
- C. Semilunar line
- D. Arcuate line
- E. Tendinous intersection

Rationale: Correct: At the linea alba (midline) good for exploratory laparotomy; advantage is that no blood vessels cross the area, good suture repair.

Incorrect: At semilunaris – incisions are rarely made due to disadvantages such disruption of the innervation. This would not separate the two rectus abdominis muscle. Pfannenstiel (arcuate line)- Above the pubic symphysis (bikini line through external abdominal oblique and rectus abdominis muscles).

Question #: 4

A 55-year-old man comes to the clinic with a bulge and discomfort in his groin. A diagnosis of indirect inguinal hernia is made, and he is admitted to the hospital. During an open hernioplasty, the spermatic cord is identified. Which of the following nerves provides innervation to component of the spermatic cord is derived from the internal abdominal oblique muscle?

- A. Femoral branch of the genitofemoral
- B. Ilioinguinal
- C. Iliohypogastric
- ✓D. Genital branch of genitofemoral
- E. Femoral

Rationale: Correct: The cremasteric muscle is derived from the internal abdominal oblique layer. It is innervated by the genital branch of the Genitofemoral (L1,2) nerve.

Incorrect: The femoral branch of the genitofemoral nerve supplies skin of the femoral triangle.

The ilioinguinal nerve innervates the Internal oblique-transversus muscles, skin over upper medial thigh, anterior scrotum and labia majora. While Iliohypogastric innervates internal oblique-transversus muscles, skin above pubis (suprapubic) and superior lateral aspect of the gluteal region. Femoral innervates skin of the anterior thigh, medial part of leg, hip and knee joints. Also the muscles of the anterior compartment of the thigh.

Question #: 5

A group of four 3rd year medical students are currently rotating in surgery. The general surgeon reviews their knowledge of the abdominal wall before having them observe a cholecystectomy. During a discussion on the blood supply, the general surgeon asks ; "Which vessel most likely gives rise to the inferior epigastric artery?"

- ✓A. External iliac
- B. Internal iliac
- C. Femoral
- D. Abdominal aorta
- E. Deep circumflex iliac

Rationale: The inferior epigastric artery, a key vessel supplying the abdominal wall, originates from the external iliac artery. Knowledge of this is crucial in surgical planning and avoiding vascular injury during procedures, hence the surgeon's question for the medical students.

Question #: 6

A 25-year-old woman comes to the physician because of a 5-month history of pain and tingling in the upper thigh. Physical examination shows decreased sensation in the medial aspect of her thigh and difficulty adducting her thigh. Imaging studies show an enlarged ovary, compressing a nerve on the lateral wall of the pelvis. Which of the following muscles is most likely innervated by this nerve?

- A. Obturator internus
- ✓B. Obturator externus
- C. Gluteus maximus
- D. Piriformis

E. Levator ani

Rationale: The patient has lost sensation in the medial thigh and has difficulty adducting the thigh. This indicates that the nerve that innervates the muscles of the medial compartment of the thigh is affected. This nerve is the obturator nerve. The obturator externus muscle is innervated by the obturator nerve.

Question #: 7

A 59-year-old man comes to the physician because of a 4-hour history of sharp pain and vesicular lesions over his left thoracic wall extending to the left nipple line. He has a history of chickenpox infection in childhood. Which of the following structures most likely contains the latent virus that has reactivated and is responsible for this patient's condition?

- A. T7 dorsal root ganglion
- B. T10 dorsal root ganglion
- C. T7 ventral root
- D. T4 ventral root
- ✓E. T4 dorsal root ganglion

Rationale: The viral particles of herpes zoster (chicken pox) can be found in the dorsal root ganglion. These can be reactivated in older patients and present as shingles.

Question #: 8

A 34-year-old woman comes to the physician for a routine pelvic examination. The physician enquires about the muscles that exit the pelvis through the greater sciatic foramen from a group of medical students. Which of the following muscles is most likely being asked for by the physician?

- ✓A. Piriformis
- B. Coccygeus
- C. Pubococcygeus
- D. Obturator internus
- E. Psoas major

Rationale: The greater sciatic foramen lies superior to the sacrospinous ligament and the ischial spine. The piriformis muscle forms the posterior wall of the pelvic cavity. It passes through the greater sciatic foramen to insert on the greater trochanter of the femur.

Question #: 9

A 45-year-old man comes to the emergency department because of 2-week history of groin pain and a palpable mass just superior to the inguinal ligament. He is diagnosed with a hernia and is prepped for surgery. During the operation, intestines are found in the deep inguinal ring. Which of the following is most likely the hernia found in this patient?

- A. Direct inguinal
- B. Umbilical
- C. Femoral
- D. Lumbar
- ✓E. Indirect inguinal

Rationale: In indirect inguinal hernias (which are located lateral to the inferior epigastric artery), intestinal structures from the abdomen pass through the deep inguinal ring and enters the inguinal canal. If the hernia enlarges, it can pass through the inguinal canal, through the superficial ring and enter into the scrotum.

Question #: 10

A 55-year-old-man is brought to the emergency department by paramedics after he was involved in a motor vehicle collision. His blood pressure is 100/65mmHg, pulse is 92/min and respirations are 17/min. Physical examination shows pink and moist mucous membranes. He has bruising and tenderness to the left upper quadrant and suprapubic region of his abdomen. During surgery a paramedian incision is attempted with medial retraction of the rectus abdominis muscle. This results in the rupture of the inferior epigastric artery. A branch of which of the following vessels will most likely secure collateral circulation to the muscle?

- ✓A. Internal thoracic
- B. External iliac
- C. Superficial epigastric
- D. Periumbilical
- E. Intercostal

Rationale: The internal thoracic artery which comes from the subclavian runs parallel to the sternum and gives off the musculophrenic and the superior epigastric vessels as terminal branches. It is the superior epigastric which anastomoses with the inferior epigastric posterior to the rectus abdominis muscle providing it with oxygenated blood.

Question #: 11

A 1-month-old boy is brought to the physician by his mother because of a 2-week history of a swelling in his right scrotum. Transillumination shows no masses but a fluid-filled space. A diagnosis of hydrocele is made. The accumulation of fluid is in the tunica vaginalis. Which of the following abdominal layers most likely gives rise to this structure?

- A. External oblique muscles
- B. Internal obliques aponeurosis
- C. Transversus abdominus muscle
- D. Transversalis fascia
- ✓E. Parietal peritoneum

Rationale: In hydrocele, the fluid accumulates in a patent processus vaginalis, which becomes the tunica vaginalis. This is formed by the parietal peritoneum that precedes the testes as they descend.

Question #: 12

A 61-year-old man comes to the physician because of a 2-month history of weakness in his right knee. Physical examination shows weakness of knee extension. Palpation of the abdominal wall in the lower right quadrant shows a firm mass verified by imaging to be located in the right iliac fossa on the lateral border of the psoas major muscle. Given the location and the symptoms, which of the following nerves in the lumbar plexus is most likely involved?

- A. Obturator
- B. Lumbosacral trunk
- C. Iliohypogastric
- D. Ilioinguinal
- ✓E. Femoral

Rationale: The femoral nerve is a branch of the lumbar plexus. It is located just lateral to the distal end of the psoas major muscle. It innervates the muscles of the anterior compartment of the thigh, which is responsible for extension of the knee joint and flexion of the hip joint.

Question #: 13

A 45-year-old woman comes to the physician because of a 1-week history of bowel incontinence. Injury to which of the following muscles is most likely responsible for her

condition?

- ✓A. Pubococcygeus
- B. Ilicoccygeus
- C. Coccygeus
- D. Obturator internus
- E. Piriformis

Rationale: The puborectalis part of the pubococcygeus muscle is the part of the levator ani that directly surrounds and supports the rectum. Therefore, if there is damage to this muscle, the patient will have fecal incontinence.

Question #: 14

A 35-year-old woman comes to the physician because of a 1-week history of a lump in her left breast. Physical examination shows a 1-cm mass in the upper outer quadrant of her left breast. Examination of A a biopsy specimen of the mass shows cancerous changes. To which of the following groups of lymph nodes would the cancer most likely metastasize?

- ✓A. Anterior axillary (pectoral)
- B. Lateral axillary (humeral)
- C. Parasternal
- D. Central axillary
- E. Apical (infraclavicular)

Rationale: The anterior axillary nodes receive up to 75% of the lymphatic fluid from the breast including the upper outer quadrant.

Question #: 15

A 25-year-old man is brought to the emergency department by his friends because of a 3-hour history of severe pain in the perineum. Physical examination shows a tender, erythematous swelling near the anal opening. A diagnosis of an anal abscess is made. During surgical debridement of the abscess, the physician is careful not to damage branches of the peripheral nerve that supply the skin and muscles of the perineum. Which of the following nerve fibers are most likely found in this nerve?

- A. Somatic afferents, visceral afferents and preganglionic parasympathetics
- B. Somatic efferents, somatic afferents and preganglionic sympathetic

- C. Preganglionic sympathetics, postganglionic sympathetics and somatic afferents
- ✓D. Postganglionic sympathetics, somatic afferents and somatic efferents
- E. Postganglionic parasympathetics, somatic afferents and somatic efferents

Rationale: The nerve that innervates the majority of the perineum is the pudendal nerve. The fibers that can be found are somatic afferents for somatic sensation, somatic efferents for skeletal muscle movement and post ganglionic sympathetic fibers to the sweat glands and arrector pill muscle in the skin.

Question #: 16

A 21-year-old woman who is a university student is suspected of having meningitis. Since this condition may alter the chemical composition of cerebrospinal fluid, a lumbar puncture is performed. Between which two vertebrae is the most appropriate site to insert the needle in this patient?

- A. T11-T12
- B. T12-L1
- C. L1-L2
- D. L2-L3
- ✓E. L3-L4

Rationale: In an adult, the spinal cord ends at the L1/L2 level. Therefore, in order to avoid damaging the spinal cord when performing a lumbar puncture, your needle needs to be inserted below the L2/L3 level.

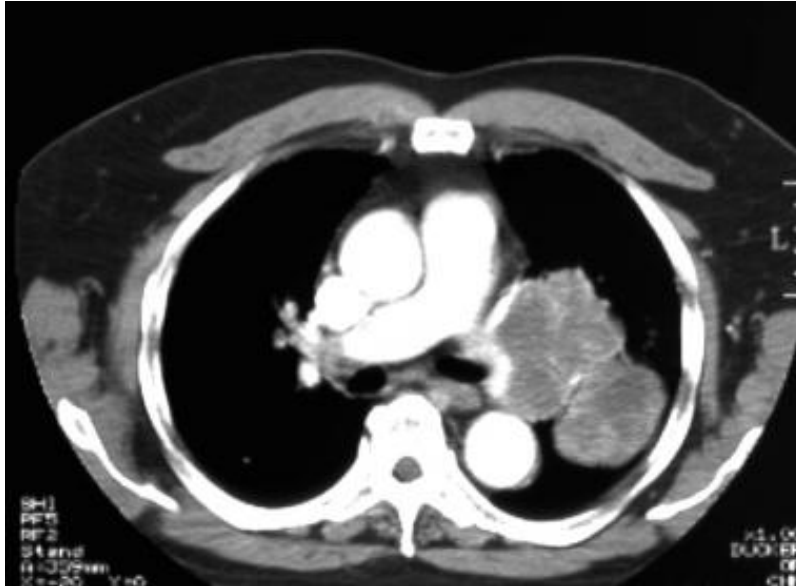
Question #: 17

A 50-year-old man is brought to the physician because of a 6-month history of cough, chest pain and unintentional weight loss. The physician orders several imaging studies, one of which is shown. This is an example of which of the following imaging modalities?

- A. Radiography
- B. Magnetic resonance imaging
- ✓C. Computer tomography
- D. Ultrasonography
- E. Angiography

Rationale: A CT scan provides detailed images of the body's structures. Given this patient's symptoms of persistent cough, chest pain, and weight loss, which could suggest a serious condition like lung cancer, a CT scan is an appropriate choice. It helps identify abnormalities such as tumors and aids in treatment planning, making it a crucial tool in this scenario.

Attachment:



Lecture 77.jpg

Question #: 18

A 54-year-old man comes to the physician because of 3-month history of weakness in his fingers. Physical examination shows difficulty flexing the distal interphalangeal joints of the index and middle fingers. Pronation and wrist flexion are weakened. There are no sensory deficits. Which of the following nerves is most likely affected?

- A. Ulnar nerve at Guyon's canal
- ✓B. Anterior interosseous nerve between the two heads of pronator teres
- C. Median nerve in the carpal tunnel
- D. Median nerve at the supracondylar ridge
- E. Posterior interosseous nerve beneath the supinator

Rationale: The AIN is a branch of the median nerve, and it primarily provides motor innervation to the deep muscles of the anterior forearm, including the flexor pollicis longus and the radial half of the flexor digitorum profundus, which flex the thumb and index finger respectively. Therefore, an issue with the AIN can lead to the described symptoms.

Question #: 19

A 78-year old woman comes to the physician because of a 1-month history of worsening stiffness of her fingers. During the physical examination, the physician holds the four fingers

of her hand together and asks her to spread her fingers. Which of the following muscles is most likely being tested??

- A. Lumbricals
- B. Palmar interossei
- ✓C. Dorsal interossei
- D. Flexor pollicis brevis
- E. Abductor pollicis brevis

Rationale: The dorsal interossei are responsible for abducting the fingers, which is tested by asking the patient to spread their fingers.

Question #: 20

A 32-year-old woman comes to the physician because of a 1-day history of pain and swelling of her pinky finger. Three days ago, she sustained a cut to her finger while chopping vegetables. Physical examination shows an erythematous pinky finger. Which of the following groups of axillary lymph nodes will most likely first receive drainage from the site of the infection?

- A. Apical
- B. Anterior (pectoral)
- C. Posterior (scapular)
- ✓D. Lateral (humeral)
- E. Central

Rationale: The humeral nodes drain the medial side of the upper limb, which is where the pinky finger is located. The apical nodes drain the lateral side of the upper limb.

Question #: 21

A 15-year-old girl is brought to the emergency department by her parents because of a 3-hour history of falling on her outstretched hand. Physical examination shows tenderness on palpation of the anatomical snuff box. An x-ray of her hand shows no obvious fractures however, the physician treats the patient by splinting the area. Which of the following bones is most likely fractured?

- A. Lunate
- ✓B. Scaphoid

- C. Hamate
- D. 1st metacarpal
- E. Triquetrum

Rationale: The floor of the anatomical snuff box is formed by the scaphoid bone. Any pain in the snuffbox is indicative of a scaphoid fracture.

Question #: 22

A 36-year-old woman comes to the physician because of a 2-month history of difficulty moving her upper limb. MRI of the shoulder shows a tear in the supraspinatus muscle. Which of the following movements of the shoulder joint is most likely affected?

- A. Flexion
- B. Adduction
- C. Lateral rotation
- ✓D. Abduction
- E. Medial rotation

Rationale: The supraspinatus muscle moves the shoulder joint from 0 to 15 degrees of abduction. It is innervated by the suprascapular nerve.

Question #: 23

A 25-year-old man comes to the physician because of a 2-month history of pain in his elbow while attempting to swing his golf club at his weekly game. Physical examination shows tenderness over the medial epicondyle. Which of the following structures is most likely inflamed?

- A. Common extensor tendon
- B. Transverse carpal ligament
- C. Palmar aponeurosis
- ✓D. Common flexor tendon
- E. Bicipital aponeurosis

Rationale: The muscles of the superficial layer of the anterior compartment of the forearm originate from the medial epicondyle from a common tendon. Inflammation of this tendon leads to medially epicondylitis, which is also called Golfer's elbow.

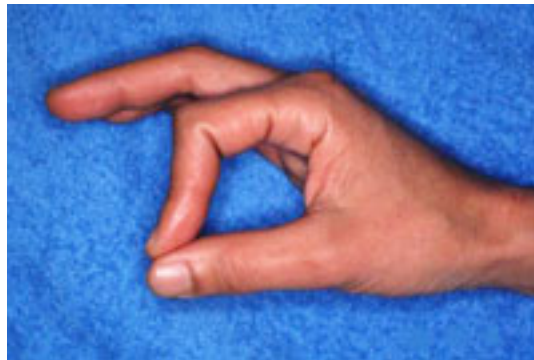
Question #: 24

A 16-year-old girl is brought to the physician by her mother because of a 2-month history of difficulty holding her pen between her thumb and index finger. Physical examination shows no sensory deficit and when asked to make the "OK" sign, her fingers are positioned as shown in the attached image. Her symptoms are most likely caused by a loss of innervation to which of the following pairs of muscles?

- A. Flexor digitorum superficialis and flexor pollicis brevis
- B. Extensor pollicis longus and extensor pollicis brevis
- C. Dorsal interossei and palmar interossei
- D. Adductor pollicis and abductor pollicis longus
- ✓E. Flexor digitorum profundus and flexor pollicis longus

Rationale: When making the OK sign, there is flexion of the MCP, PIP and DIP of the index finger and flexion of the MCP and IP joints of the thumb. In the image, the DIP of the index finger is extended, the MCP of the thumb is extended and there is only slight flexion of the IP joint of the thumb. This indicates that the muscle that flexes the DIP of the thumb, the flexor digitorum profundus and the muscle that flexes the MCP of thumb, flexor pollicis longus, are not functioning. These muscles are located in the deep compartment of the anterior forearm and are innervated by the anterior interosseous nerve. There is still slight flexion of the IP joint of the thumb, because the flexor pollicis brevis which is innervated by the recurrent median nerve is unaffected.

Attachment:



anterior_interosseous_nerve_palsy_01.jpg

Question #: 25

A 27-year-old man is brought to the emergency department because of an injured left arm after losing control of his motorcycle. Physical examination shows an inability to abduct his arm to the horizontal, and a diagnosis of axillary nerve injury is made. This injury will most likely result in sensory findings in which of the following areas?

- ✓A. Upper lateral aspect of the arm
- B. Upper medial aspect of the arm
- C. Lateral aspect of the forearm
- D. Medial aspect of the forearm
- E. Posterior aspect of the arm

Rationale: Correct: Superior lateral brachial cutaneous nerve (branch of Axillary nerve) supplies upper lateral aspect of the arm.

Incorrect: Inferior lateral brachial cutaneous nerve(branch of Radial nerve) supplies lower lateral aspect of the arm. Whereas upper medial aspect of the arm is supplied by Medial brachial cutaneous nerve and Intercostobrachial nerve (T2).

Question #: 26

A 68-year-old man comes to the physician because of a history of back pain that irradiates to his right thigh. He is diagnosed with a posterior abdominal wall abscess which affects the iliopsoas muscle. Which of the following functions will most likely be affected in this patient?

- A. Medial rotation of the thigh
- B. Extension of the thigh
- C. Flexion at the knee
- D. Lateral rotation of the thigh
- ✓E. Flexion of the thigh

Rationale: Iliopsoas muscle is a flexor of the hip and trunk and lateral flexor of the trunk.

Question #: 27

A 40-year-old man comes to the physician for a follow-up examination because of ambulation difficulty following a motor vehicle collision. Physical examination shows he falls towards his right side when attempting to stand on his left foot. Which of the following nerves has most likely been injured?

- A. Left obturator
- B. Left inferior gluteal nerve
- C. Right inferior gluteal nerve

✓D. Left superior gluteal nerve

E. Right superior gluteal nerve

Rationale: Superior gluteal neuropathy can be the result of nerve compression of the superior gluteal nerve. Clinical presentation includes weakness in hip abduction and external rotation with limping gait. Clinical examination will reveal a positive Trendelenburg sign with a pelvic tilt towards the strong side.

Question #: 28

A 36-year-old man comes to the emergency department because of a 2-day history of pain and tenderness in his right wrist after a fall on his outstretched hand. Physical examination shows tenderness on the lateral side of the wrist, just proximal to the base of the first metacarpal. Which of the following is the most likely diagnosis?

A. Fracture of the fifth metacarpal

B. Fracture of the hamate

C. Tenosynovitis of thumb extensors

✓D. Fracture of the scaphoid

E. Tenosynovitis of the long flexor tendons

Rationale: The scaphoid bone, which is just proximal to the base of the first metacarpal, forms the floor of the anatomical snuff box. Tenderness in this area indicates that there is a fracture of the scaphoid bone.

Question #: 29

A 60-year-old multiparous woman comes to the physician because of a 6-month history of episodes of fecal incontinence. This is due to weakness in one of the muscles of the pelvis. A tear in which of the following muscles would most likely result in this patient's symptoms?

A. Iliococcygeus

B. Piriformis

C. Obturator internus

D. Coccygeus

✓E. Puborectalis

Rationale: Puborectalis muscle is part of levator ani group of muscles. Puborectalis takes attachment from the pelvic surfaces of both ischiopubic rami, anterolaterally. It passes posterior to the rectum to form a muscular sling. Therefore weakness of it can lead to fecal incontinence.

Question #: 30

A 45-year-old man comes to the clinic because of a 2-week history of a bulge in his right groin. He reports that the bulge increases in size when standing or coughing but reduces when lying down. His temperature is 36.8°C (98.2°F), pulse is 76/min, respirations are 18/min, and blood pressure is 128/82 mm Hg. Physical examination shows a soft, non-tender, reducible mass in the right inguinal region that is lateral to the inferior epigastric vessels. Which of the following is the most likely diagnosis?

- A. Direct inguinal hernia
- ✓B. Indirect inguinal hernia
- C. Femoral hernia
- D. Spigelian hernia
- E. Obturator hernia

Rationale: Indirect inguinal hernias occur lateral to the inferior epigastric vessels, protruding through the deep inguinal ring and potentially into the scrotum